

(Viscosity of Liquids)

Pre-Lab Plan
Unit Operations Lab 2
9/17/2019

Group 1, Tuesday, AM section

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1. Experimental Objective:

The purpose of this experiment is to understand how to calculate viscosity using a viscometer. This involves two parts. The first part is to determine the viscosity of Newtonian fluid using Hagen-Poiseuille equation

$$Q = \frac{\Delta P \pi R^4}{8 \mu L}$$

The second part is to determine the viscosity of Non-Newtonian fluid using equation

$$\frac{m}{n} \left[\frac{3n+1}{n} \right]^n \cdot \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} R \right)$$

2. Experimental:

2.1 Apparatus

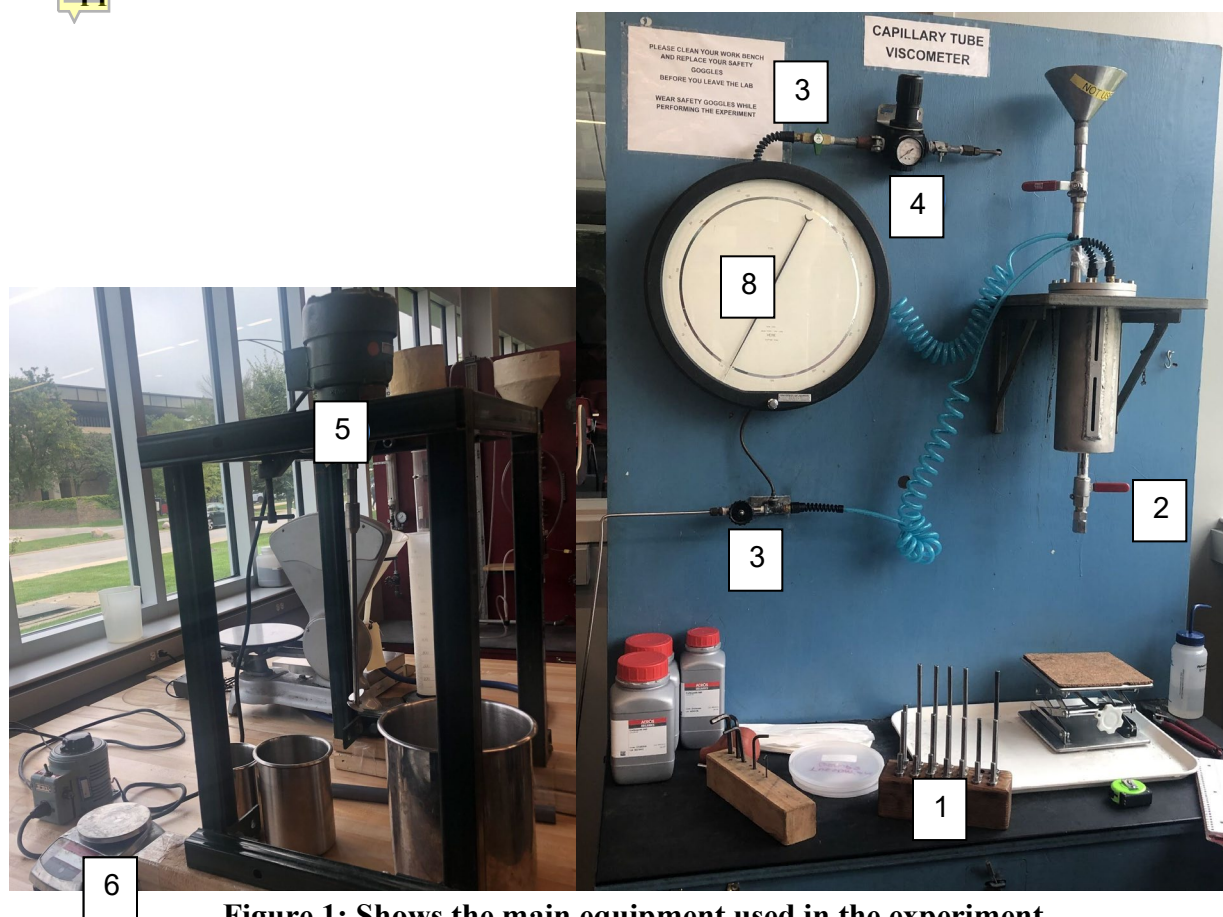


Figure 1: Shows the main equipment used in the experiment

Table 1: The important components used in the experiment

#	Important components	Description
1	Capillary tube	Used to calculate the viscosity
2	Pressurization vessel	Applies pressure to capillary tube
3	Vent valve	To pressurized/depressurized the vessel
4	pressure regulator	To adjust the pressure applied to vessel
5	Power mixer	Used to prepare the Carbopol solution
6	Mass scale	Used to determine mass
7	Pressure Gauge	To read the pressure being applied to tube

2.2 Materials and Supplies:

Table 2: Materials and Supplies used in the experiment

#	Material	Description
1	Air	Used to pressurized the fluid
2	Warm water	Mixed with Carbopol to prepare non-newtonian fluid
3	Carbopol	Used to create non-newtonian fluid
4	NaOH	To neutralized the obtained product
5	pH indicator	To measure the pH of the obtained solution

2.3 Experimental plan:

Preparation of the Carbopol Solution

1. Prepare approximately 2 liters of 0.8% wt Carbopol solution
2. Insert the blade $\frac{1}{4}$ from the bottom of the container to not damage the blade
3. Add 12g of Carbopol in small amounts to constantly agitated 1.5L of warm water to achieve 0.8%.
4. Since Carbopol is naturally acidic, use a neutralizing agent to achieve the ideal pH of 6.5-7.5 in order to reach optimum viscosity

Capillary Tube runs

1. Make several runs of different capillary tubes of different diameter and length. Collect the flow rate at 4-5 different pressure differentials. Max pressure is 60 psig
2. Find the time to flow through each capillary tube and the volume of the Carbopol that flowed through
3. For each capillary tube measure the pressure at 10, 25, 35, 45 psig
4. Use the capillary tubes of length 3, 6, and 9 in. For each of those use ones of diameter 0.062 in and 0.046 in.
5. For each of the runs use a stopwatch from the time the run is started all of the Carbopol exiting to calculate the volume flow rate.
6. Use a mass balance to find the amount of carbopol that is exiting and then convert to volume using density.



Project Safety Evaluation (10 pts)

Project Title: Viscosity of Liquids

Date: 9/17/2019

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Power outlet must be kept dry
Potential mechanical hazards?	Yes	If yes, identify conditions: Power Mixer must be handled Properly
Potential pressure hazards?	Yes	If yes, identify conditions: Careful
Potential temperature hazards?	NO	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: NaOH do not ingest any chemical or touch without gloves
Gloves required?	Yes	If yes, specify type: Chemical resistant Rubber
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Carbol/P01 940, NaOH
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	25	
Pressure (psia)	60	Careful when removing capillary tubes make sure
Safety Controls	yes/no	If yes, Provide description
	Yes	Power Mixer must be handled properly

I have read relevant background material for the Unit Operations Laboratory entitled: "Viscosity of Liquid" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

 9/17
signature of team member date

 9/17
signature of team member date

 9/17
signature of team member date

